BCS-505	Introduction to Microprocessor	L T P 3 0 0	3 Credits

Pre-requisites: Basic computer knowledge

Course Objectives:

CO1	To introduce students with the architecture and operation of typical microprocessors and microcontrollers
CO2	Know the various functional units of m.icroprocessor
CO3	To familiarize the students with the programming and interfacing of microprocessors and microcontrollers.
CO4	To provide strong foundation for designing real world applications using microprocessors and microcontrollers.
CO5	Know the various peripheral devices.

Detailed Syllabus

Module 1

Introduction : Microprocessor evolution and types, microprocessor architecture and operation of its components, addressing modes, interrupts, data transfer schemes, instruction and data flow, timer and timing diagram. Interfacing devices. Architectural advancement of microprocessor. Typical microprocessor development schemes.

Module 2

8-bit Microprocessors: Pin diagram and internal architecture of 8085 microprocessor, registers, ALU, Control & statu, interrupt and machine cycle. Instruction sets. Addressing modes. Instruction formats Instruction Classification: data transfer, arithmetic operations, logical operations, branching operations, machine control and assembler directives.

16-bit Microprocessor: Architecture of 8086 microprocessor: register organization, bus interface unit, execution unit, memory addressing, memory segmentation. Operating modes. Instruction sets, instruction format, Types of instructions. Interrupts: hardware and software interrupts

Module 3

Programming: Assembly language programming based on intel 8085/8086. Instructions, data transfer, arithmetic, logic, branch operations, looping, counting, indexing, programming techniques, counters and time delays, stacks and subroutines, conditional call and return instructions

Peripheral Interfacing: Peripheral Devices: 8237 DMA Controller, 8255 programmable peripheral interface, 8253/8254programmable timer/counter, 8259 programmable interrupt controller, 8251 USART and RS232C

Text/Reference books:

1. Gaonkar , Ramesh S , "Microprocessor Architecture, Programming and Applications with 8085", Penram International Publishing.
2. Ray A K , Bhurchandi K M , "Advanced Microprocessors and Peripherals", TMH
3. Hall D V ,"Microprocessor Interfacing', TMH
4. Liu and Gibson G A , " Microcomputer System: The 8086/8088 family'' ,PHI
5. Aditya P Mathur, " Introduction to Microprocessor", TMH
6. Brey, Barry B, "INTEL Microprocessors", PHI
7. Renu Sigh & B.P.Sigh, "Microprocessor, Interfacing and Applications
8. M Rafiqzzaman, "Microprocessors, Theory and Applications",

CO1	Recall and apply a basic concept of digital fundamentals to Microprocessor based personal computer system.
CO2	Identify a detailed s/w & h/w structure of the Microprocessor.
CO3	Illustrate how the different peripherals (8255, 8253 etc.) Are interfaced with microprocessor.
CO4	Analyze the data transfer information through serial & parallel ports.
CO5	Train their practical knowledge through laboratory experiments.